

Effects of Explainable Artificial Intelligence on trust and human behavior in a high-risk decision task

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Abstract

Understanding the recommendation of an artificially intelligent (AI) assistant for decision-making is especially important in high-risk tasks, such as deciding whether a mushroom is edible or poisonous. To foster user understanding and appropriate trust in such systems, we tested the effects of explainable artificial intelligence (XAI) methods and an educational intervention on AI-assisted decision-making behavior in a 2x2 between subjects online experiment with $N = 410$ participants. We developed a novel use case in which users go on a virtual mushroom hunt and are tasked with picking only edible mushrooms but leaving poisonous ones. Additionally, users were provided with an AI-based app that shows classification results of mushroom images. For the manipulation of explainability, one subgroup additionally received attribution-based and example-based explanations of the AI's predictions, and for the educational intervention one subgroup received additional information on how the AI worked. We found that the group with explanations outperformed the group without explanations and showed more appropriate trust levels. Contrary to our expectation, we did not find effects for the educational intervention, domain-specific knowledge, or AI knowledge on performance. We discuss practical implications and introduce the mushroom-picking task as a promising use case for XAI research.

1. Introduction

Imagine a tourist on vacation in Austria joining locals in mushroom picking to explore the country's nature. Since the tourist has not previously gone mushroom picking in Austria, they decide that some kind of classification aid would be helpful. The obvious choice would be to use a mushroom classification book. Due to the vast variety of mushrooms that grow in Austria, however, the tourist decides to use an app that is based on an artificial intelligence (AI) and aims to identify edible and inedible/poisonous mushrooms from images¹. Discriminating mushrooms is a challenging task, as many poisonous mushrooms closely resemble edible ones. If a person is not knowledgeable about Austrian mushrooms, they might easily pick an inedible mushroom that was (for whatever reason) incorrectly classified as edible by the AI-driven app. In such a scenario the user trusts the system too much (*overtrust*), which results in a potentially dangerous error.

Image classification has long been a standard use case of AI (Lu & Weng, 2007; Rawat & Wang, 2017). While classification errors may be harmless in some contexts—such as the animal identification examples that are often used

as a “proof of concept”—, such errors can have serious consequences in high-stakes domains such as mushroom picking (see reports of mushroom poisoning, e.g., Brandenburg and Ward, 2018) or medical image analysis (Jiao et al., 2021; Litjens et al., 2017). In order to avoid errors in AI-assisted decision-making tasks (i.e., too low acceptance of reliable classification systems and too high acceptance of wrong AI classifications) humans need to understand how an AI arrives at a particular decision and must be able to evaluate whether a decision is reasonable or seems odd (e.g., Körber et al., 2018; Kraus et al., 2020; Long & Magerko, 2020; Parasuraman & Riley, 1997). It is thus important to understand the system's functionality and limitations for trust calibration. Trust evolves and adapts over time with growing knowledge until, ideally, “a user's trust in a system corresponds exactly with its objective capabilities and its actual performance” (Kraus et al., 2020, p. 719). Such a level of system comprehension can be achieved by (i) enabling AI systems to explain their decisions to users, which is often referred to as *explainable AI* or *XAI* (see e.g., Long & Magerko, 2020; Miller, 2019); or (ii) improving humans' AI literacy such that they better comprehend AI systems, their decisions, and limitations (see e.g., Long & Magerko, 2020; Moehring et al., 2016; Ng et al., 2021). This requires research into user comprehension and trust of AI-based classification systems—especially for high-risk decision tasks—and how interventions such as user-centered system design and user training can affect these criteria of comprehension and trust.

Psychological research can help in this endeavor from two prominent angles and research traditions: *engineering psychology*, which is concerned with adequacy of system design (e.g., Carayon, 2006); and *psychology of learning and individual differences*, which is concerned with human knowledge, comprehension and problem-solving (e.g., Greiff

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¹We would like to emphasize that this is only an imaginary example for the purpose of illustration. The safe identification of mushrooms is a difficult task that requires a lot of expertise and training. Professional mycological associations therefore expressly warn to go mushroom picking without necessary expertise.

et al., 2014; Moehring et al., 2016) and how these can be improved effectively by educational interventions (Pangrazio et al., 2020). Building on these research traditions, this article presents the findings of an exploratory study of the interaction between human users and an AI-based classification system in a high-risk decision task. In an experiment, we explored the effect of an educational intervention regarding how AI systems work (AI literacy research tradition) and the effect of visual explanations (XAI research tradition) on user decision-making, comprehension, and trust. Our study had the following objectives:

1. Explore the effects of an XAI interface with attribution-based and example-based visual explanations by comparing it with an interface that does not explain its recommendations.
2. Explore the effects of an educational intervention presented as texts and illustrations about how machine learning works by comparing two groups—one with and one without such an intervention.
3. Understand the underlying psychological mechanisms by assessing users' prior knowledge of AI, trust, and domain-specific knowledge as major factors commonly used to explain differences in comprehension and system use.
4. Introduce mushroom picking as a new use case that can be employed for psychological research into comprehension and trust in AI-based classification systems as well as decision-making in high-risk contexts.

2. Theory and related work

As noted in the introduction (Section 1), this article is concerned with researching user performance in AI-assisted decision tasks by intervening on user comprehension of the AI system. We do this coming from two different directions: Educating the user on the one hand and enabling systems to explain their decisions on the other hand.

Therefore, the first part of the theory section is concerned with user comprehension. We describe how domain-specific knowledge and knowledge about the technology being used to solve problems are important for task success by discussing literacy constructs. We sketch how knowledge about AI technology could consequently be tackled by educational interventions.

In the second part, we describe how user understanding during human–technology interaction can be improved by interface design. In doing so, we discuss recent developments in visual explanations for AI-based decisions that aim to open up so-called “black-box” models.

Finally, as trust is a key construct in understanding the dynamics in human decision-making with AI-assisted decision aids we describe how comprehension and trust are related, and consequently should be affected by both, educative and system-design interventions.

2.1. Comprehending system outcomes: Improving users' AI literacy

As noted in the introduction (Section 1), one possible intervention from a human factors perspective is to educate users to better comprehend AI technology (i.e. educational interventions). This general ability of users to comprehend AI technology and the ability to use this sort of technology is often labeled *AI literacy*. While AI literacy is hard to define, it can be described as a complex construct based on the integration of different competencies, including for example to know and understand AI (e.g., know basic AI functions) (Long & Magerko, 2020; Ng et al., 2021).

Literacy constructs have not only been used with respect to AI. Several similar skill sets and closely related competencies in combination with (information) technology are conceptualized as partly overlapping literacies. Examples are information and communication technology (ICT) literacy (e.g., “the ability to understand and work with modern information technology, such as the Internet” Moehring et al., 2016, p. 171), digital literacy (e.g., “skill, knowledge, ethics, and creative outputs in the digital network environment” Covello and Lei, 2010, p. 3), or information literacy (e.g., “the ability to find sources, analyze, and synthesize the material, and evaluate the source credibility” Moehring et al., 2016, p. 171). Due to overlaps, multitude of labels, and imprecise definitions, it is difficult to describe how these constructs may relate to each other. For example, while for Long and Magerko (2020) digital literacy is the prerequisite for AI literacy, Moehring et al. (2016) interpret digital literacy more as an “overarching concept encompassing several disciplines” (p. 171 Moehring et al., 2016).

However, what these constructs have in common is their overlap with traditional comprehension abilities and the associated cognitive requirements (Moehring et al., 2016). The traditional comprehension construct can be defined as “the process of developing mental representations, by which prior long-term knowledge is incorporated with the available information given through text, audio, or video through complex mental processes” (Moehring et al., 2016, p. 171).

The literature showed that comprehension ability can be explained to a large extent by underlying cognitive abilities such as fluid intelligence or domain-specific knowledge, and as these constructs are independent of presentation of information, comprehension can be viewed as unidimensional. For example, Schroeders et al. (2010) report that listening, reading and viewing comprehension are highly correlated and all three are largely influenced by crystallized intelligence (measured as declarative knowledge items). This can be explained by the theoretical assumption that knowledge facilitates information retrieval (Moehring et al., 2016). It is thus reasonable that such traditional comprehension constructs can also be extended to other literacies that require users to search and integrate necessary information in order to answer test questions or make decisions.

For example, Moehring et al. (2016) tested if fluid intelligence and crystallized intelligence—or, more concretely, domain-specific knowledge—can explain variance in an

ecological momentary assessment of digital literacy where information had to be retrieved from the internet—a task requiring more complex processes such as to evaluate, weight and integrate potentially contradicting information or to evaluate the credibility of a source of information (Moehring et al., 2016). Furthermore, it is conceivable that participants' computer usage could also influence digital literacy. However, results show, that computer usage had only a negligibly small influence on the comprehension construct, but variance was largely explained by domain-specific knowledge and fluid intelligence (Moehring et al., 2016). In contrast to this, based on work by Greiff et al. (2014) it could be argued that the influence of such technology-related competencies or knowledge could vary depending on the level of complexity of human–computer interaction requirements of the task (Greiff et al., 2014).

Similarly, by analyzing the definitions of AI literacy (Long & Magerko, 2020; Ng et al., 2021) and its theoretical relations to other literacy constructs, it could be assumed that AI literacy can also be explained by domain-specific knowledge (i.e., mushroom knowledge when a mushroom classifier is used) and fluid intelligence. However, the use of artificial intelligence as part of problem solving or decisions of picking the right option (e.g., declaring a mushroom as edible when it is in fact edible) also differs from traditional comprehension tests. While searching for information on the internet, as shown by Moehring et al. (2016), required additional cognitive processes such as evaluating the credibility of the source or integrating information from multiple documents, the task of correctly identifying mushrooms as edible, inedible, or poisonous with an AI-based assistant system would also require the evaluation of the system's outcome or the credibility of the system as an independent source of information in general. It is thus conceivable that users need a certain degree of AI background knowledge and knowledge about the system's functionality. This could mean that system knowledge, as a facet of AI literacy, would be a variable (and a precondition) for system comprehension in interaction situations.

Especially in the human factors literature, the need for appropriate knowledge and understanding concerning a new technology such as assistant systems or autonomous driven cars are essential for the development of trust and appropriate usage of the system (see Körber et al., 2018; Sauer et al., 2016). For example, Körber et al. (2018) show that “introductory information” on an automated vehicle before a driving task had medium effect on the participants trust levels and reliance behavior which resulted in lower collision rates with an obstacle for an informed group of participants during a driving simulator study.

2.2. Explainable AI: Opening the black box

Besides educational attempts to improve AI literacy, another option to establish an adequate amount of system comprehension is the human-centered design of AI systems that can explain their decisions to foster human comprehension of outcomes.

Most state-of-the-art machine learning models—in particular deep neural networks (DNNs)—are based on vast amounts of data and deeply nested mathematical functions. Such models are considered to be “black boxes” because it is difficult for humans to understand how and why they arrive at certain decisions. Explainable artificial intelligence (XAI) methods are commonly used to make black-box models more understandable for human users (Alicioglu & Sun, 2022; Barredo Arrieta et al., 2020; Miller, 2019). While there exist models that are intrinsically explainable (i.e., models with high degree of transparency; Barredo Arrieta et al. 2020), they often come with the drawback of producing less accurate results. Therefore, post-hoc explanation techniques are employed to explain existing models (Lipton, 2018).

There exists a large body of work on XAI, and several attempts have been made at classifying the various techniques (Du et al., 2019; Guidotti et al., 2019; Molnar, 2022; Murdoch et al., 2019). For example, techniques can be divided into global and local methods. Global explanation methods give insights about the model as a whole, whereas local methods yield insights for individual data items (i.e., instances). XAI techniques can also be distinguished based on whether they are only applicable for specific types of models (model-specific), or whether they can be applied to any model (model-agnostic).

The AI model used in our study to identify different types of mushrooms is a deep convolutional neural network (CNN). In the following we give an overview of existing domain-specific XAI techniques for CNNs, which we grouped into *feature-based*, *attribution-based*, and *example-based* techniques.

Feature-based Deep neural networks transform their input into a numerical representation (also called *features*) that are automatically derived during training. Visualizations of these features showed that early layers of image classification networks learn low-level concepts like edges or simple patterns, while later layers are specialized at detecting high-level concepts (Olah et al., 2017; Zeiler & Fergus, 2014). In a similar fashion, network dissection (Bau et al., 2020) aims to explain to which feature or concept a certain unit corresponds. In the case of a CNN, “unit” refers to a single convolution kernel, which can be roughly thought of as a pattern detector.

Attribution-based In the domain of image classification, *attribution*-based explanations are commonly used to explain which regions of an image were important for the model's decision. The most important regions are typically visualized directly within the image. Examples of such techniques are saliency maps (Simonyan et al., 2014), deconvnet (Zeiler & Fergus, 2014), Grad-CAM (Selvaraju et al., 2017), LIME (Ribeiro et al., 2016), and SHAP (Lundberg & Lee, 2017).

Example-based In this type of explanation techniques, certain *exemplary* data items are picked out and shown to users. Such exemplary instances can be prototypes (Bien & Tibshirani, 2011)—typical representatives of certain classes—or counter-examples that differ from these prototypical

instances. Nearest-neighbor techniques are used to extract training instances that are closest to a certain target data point (Cai et al., 2019; Jeyakumar et al., 2020). Prior work on example-based explanations suggests that humans prefer example-based methods over other explanations (Cai et al., 2019; Jeyakumar et al., 2020; Kim et al., 2016).

Most of these XAI techniques were originally developed by domain experts such as AI model developers in an attempt to better understand their own creations. As such, they are typically not optimized for being shown to lay end users, and knowledge about their effectiveness with this target audience is still limited.

While there already exists a number of empirical studies with human participants to test the effect of different XAI techniques and/or to test underlying processes (e.g., Bućinca et al., 2020; Cai et al., 2019; Kenny et al., 2021; van der Waa et al., 2021; Wang & Yin, 2021; Yang et al., 2020; Zhang et al., 2020), the field is only at the beginning. Bućinca et al. (2020), for instance, tested the effect of an *example-based* explanation and an *attribution-based* explanation technique using a task in which participants had to decide if pictures of food had a certain amount of fat. They found that explanations led to higher trust levels, higher ratings of understanding, and better performance (higher accuracy). However, they also showed that results are inconsistent when compared to a second experiment with different task instructions (proxy task).

Additionally, taking together results from other studies, no clear conclusions can be drawn as (i) there exist a few studies mostly from recent years only, (ii) studies show only mixed results, (iii) studies vary in quality as also criticized by Kenny et al. (2021), (iv) studies vary in contexts, such as artificial and abstract tasks like number recognition (Kenny et al., 2021), nutrition (Bućinca et al., 2020), video game play (Huber et al., 2021), and others, and (v) studies vary in study design and the constructs that have been measured. It is thus hard to draw summary conclusions—a general problem of behavioral studies as emphasized by Meehl (1990). One example of a potential moderator causing different results in the literature could be differences in AI error rates (Kenny et al., 2021), amongst many other possible candidates. Differences in study designs must not be understood as a bad thing, but they make summaries difficult, which in turn shows that we are only at an early stage of research. Additionally, generalizations are difficult and effects need to be tested for every new use case again. Clear theory and accurate predictions may not be possible to this day calling for a higher demand of exploration as a first step of theory building.

2.3. Comprehension and trust calibration in human-machine interaction

One key variable within models of human-technology interaction such as automated driving, human-robot interaction or other assistant systems—and consequently also AI-based systems—is the psychological construct of trust (see Hoff & Bashir, 2015; Körber, 2019; Lee & See, 2004; Parasuraman &

Riley, 1997). In modern literature, trust may be defined as an attitudinal construct describing the willingness of users to be vulnerable to the actions of some automated system to achieve some goal (Hannibal et al., 2021; Lee & See, 2004), and such trust may be dynamically adapted (Kraus et al., 2020). Note that a situation of vulnerability and uncertainty is key to study trust as they are a prerequisite for trust (Hannibal et al., 2021; Lee & See, 2004).

However, a high level of trust cannot be a goal in itself. Rather, an appropriate level of trust—so-called calibrated trust—should be achieved, which also corresponds to the competencies of the system (e.g., Kraus et al., 2020). System designers thus aim at developing a system which can establish an amount of trust that is neither too low nor too high. Too low trust levels may cause humans to avoid using a system despite it being actually beneficial for the task outcome (distrust). Too high trust levels may lead to errors and negative consequences when a human user trusts an automated system more than its actual competences and boundaries would allow for (overtrust). In the domain of automated driving this could mean trusting an automated system without intervening (e.g., breaking or steering manually) although a user intervention would have been necessary due to system failure and /error in order to avoid an accident (e.g., Körber et al., 2018; Kraus et al., 2020).

In our mushroom-picking example, calibrated trust means knowing in which situations and how the system can be valuable and actually deliver additional information that can be trusted, but also not falling for overtrust by picking recommended mushrooms in situations where a system failure should have been considered (e.g., knowing from low quality of the input image that the AI should not be able to classify the mushroom reliably). In this respect, it is necessary to understand how the system works and where its boundaries are (Körber et al., 2018), which can be achieved by user education (Körber et al., 2018) or by system transparency (Kraus et al., 2020).

3. Research question

Having summarized the literature in the previous section, this section will derive open questions that we will then address through exploratory experimentation.

3.1. Exploring the effects of visual explanations and an educational intervention about AI on user performance, trust and comprehension

As stated in previous sections, there are two main routes toward user comprehension of AI-based decisions to achieve improved performance and adequate trust levels: improving users' general understanding of AI systems (i.e., AI literacy), or enabling the AI system to explain its decisions (i.e., AI explainability).

The goal of this study was to test the effects of an educational intervention about AI systems, and the effect of visual explanations on user cognition and behavior—which means performance, trust, and comprehension. While there

already exist some initial studies on the effects of educational interventions in the human factors literature (Körber et al., 2018) as well as on the effects of visual explanations (e.g., Buçinca et al., 2020; Kenny et al., 2021), the overall literature is still sparse and requires further research due to mixed results and varying quality, contexts, and study design. Thus a robust prediction of effects is not possible at this point.

Due to this lack of theory and robust data, we as AI researchers might simply not be ready to test specific hypotheses for certain contexts—a conclusion that is not unheard in other research fields such as psychology (Scheel et al., 2021) or human–robot interaction (Leichtmann et al., 2022). Scheel et al. (2021) suggest that, before testing hypotheses confirmatorily, preceding steps such as exploratory experimentation can help to test relationships between variables or to explore boundaries on the basis of which more concrete theoretical assumptions can be made. Only afterwards specific hypotheses can be tested with corresponding power. Therefore, we decided to choose exploratory experimentation as the basis of inquiry in our study.

To foster the development of stronger theory, we additionally included some of the most central variables known from the psychological literature. Besides task performance as a central variable, trust in the AI system also plays an important role, especially in situations of vulnerability (Hannibal et al., 2021), as discussed in Section 2.3. Under certain circumstances, such as in high-risk contexts with uncertainty, task performance might also reflect the construct of trust in the form of reliance. This is especially evident in situations where the AI makes an incorrect recommendation and overtrust causes the user to follow it. We thus explore the effect of visual explanations and an educational intervention on task performance and trust, and the situation in which the AI classifies incorrectly is of special interest to explore overtrust.

Known from the past literature of traditional comprehension studies, but also from studies on other literacies such as ICT literacy, domain-specific knowledge is a key predictor variable. Although the literature on the influence of technology knowledge in such studies is mixed, Greiff et al. (2014) argue that the influence of such technology-specific knowledge (as included in the construct of ICT literacy) might have to be tested for every new use case. We thus include both, domain-specific knowledge and AI knowledge in our exploratory model testing.

Human behavior is determined by the context of use. Therefore, a detailed description of the use case and its characteristics is important, as this study context determines to what extent the results are generalizable to other contexts. For our exploratory experimentation we chose the use case of mushroom picking due to a number of advantages outlined in the next section.

3.2. In search of a use case: Why mushroom picking is a promising context for XAI

Researchers in the past have argued that results and recommendations for human-centered interaction with AI

might look different depending on the context of use. For example, Behnke et al. (2017) and Leichtmann et al. (2018) argue that plan linearization of an AI-based planner could look different depending on the problem space. In the area of XAI, the importance of careful selection of use cases was also emphasized, as differences in contexts affect the transferability of study results (Buçinca et al., 2020; Kenny et al., 2021).

As a consequence, in order to set the boundaries for which a transfer of empirical findings from one use case to another is possible, contextual characteristics have to be well described and defined. A generalization from specific use cases to a different one might then only be valid if it is similar in the most central characteristics. In the case of visual explanations of an AI recommendation, for example, use cases vary in the severity of consequences and thus affect different values of users. Thus, a proper description of the use case is crucial.

In order to research the effect of visual explanations of AI decisions and the effect of an educational intervention about AI on human decision-making, comprehension, and trust, we chose the use case of mushroom picking with an AI classifier for the following reasons:

Relevance of decision Use cases differ with regards to whether or not it is of a certain value for users that the outcome of an AI-assisted decision is accurate. As stated in the introduction (Section 1), a wrong classification of cat and dog pictures will have no or low impact on human lives, whereas incorrectly classified mushrooms can have severe health consequences for the user. Mushroom poisoning leads to symptoms such as vomiting, dizziness, loss of consciousness, or even death (Brandenburg & Ward, 2018). These situations of vulnerability are especially important when researching trust in human–technology interaction as vulnerability is a prerequisite for trust (Hannibal et al., 2021). By choosing the high-risk context of mushroom picking, we ensure a certain degree of relevance for users.

Availability of target population Many use cases, such as medical image analysis (e.g., Jiao et al., 2021) and detection of dangerous objects in X-ray images of luggage (e.g., Hättenschwiler et al., 2019), have a high relevance of decision, but require specifically trained experts. Such expert users are often difficult to recruit for participation in scientific studies. For quantitative research (typically requiring large sample sizes to achieve high statistical power) it might be easier to have a target population with higher availability. Since mushroom picking is part of the culture for many (European) countries (e.g., Aigner & Krisai-Greilhuber, 2016; Kaaronen, 2020; Svanberg & Lindh, 2019), we can draw participants from a larger population of people engaging in mushroom picking, or at least from a population of people who can easily imagine such a scenario.

Closeness to participants' reality Scientific studies with human participants usually have a certain artificiality. When people are tasked to imagine a certain situation, or use some technology they would not normally use, they often become

aware of the study situation. To mitigate this problem, the use case should be as close to a real user situation as possible. Participants can then easily imagine the situation and their behavior in the study should be a better estimator for real-world behavior. A use case close to participant's reality is challenging to find. Financial decisions, for instance, usually take a lot of time and critical reflection. If users are tasked to imagine how they would make a certain financial decision within a short study, their decisions might not reflect their real-world behavior. In contrast, the decision of whether to pick a mushroom or leave it is typically a fast decision made within seconds or minutes and is characterized as "fast and frugal" based on simple rules such as avoiding spotted mushrooms (Kaaronen, 2020). Therefore, although the mushroom-picking behavior is studied in an abstract research situation, it might be easier for participants to imagine going mushroom picking compared to other contexts in which decision-making is done more thoughtfully and takes more time. We thus expect the results to be closer to true values compared with results in other artificially constructed situations.

Public interest Certain use cases—like building a home theater—might be close to people's reality and easily imaginable in laboratory experiments (Leichtmann et al., 2018), but they lack public interest. While mushroom picking may not be prominent in all cultures, it is an integral part of cultural heritage in many European countries. This is indicated by studies from Sweden (Svanberg & Lindh, 2019), Finland (Kaaronen, 2020), and Austria (Aigner & Krisai-Greilhuber, 2016). As an outdoor activity, mushroom picking also has a recreational aspect (see e.g., Aigner & Krisai-Greilhuber, 2016), relating it to health-trends such as "shinrin-yoku", the Japanese practice of "forest bathing" (Park et al., 2010). Such activities are assumed to have positive effects on human health and well-being which is echoed in studies on mindfulness (Howell et al., 2011). Furthermore, mushroom picking embodies values of culture and tradition (Aigner & Krisai-Greilhuber, 2016; Peintner et al., 2013; Svanberg & Lindh, 2019). Finally, as mushroom picking is mostly associated with the intention to eat the collected mushrooms (Aigner & Krisai-Greilhuber, 2016; Kaaronen, 2020), the reliable identification of edible mushrooms is of particular interest.

Furthermore, the availability of mushroom image databases due to AI "challenges" (FGCVx Fungi Classification Challenge, 2018) and community-driven projects (Danish svampeatlas, 2022) allows to train an actual model as a basis for the AI-assisted decision-making.

4. Methods

We conducted an online study, in which participants were tasked to decide—for a set of mushroom pictures—if a depicted mushroom is edible or poisonous² and if they

²For this study, we did not distinguish between poisonous and inedible mushrooms.

would pick the mushroom for cooking or leave it. For each decision task, participants were shown a fictitious app named *Forestly*. The app aided the participants in their decisions by showing the classification results of an artificial intelligence.

In the following section, we describe the methods used for our exploratory experimentation including study design, sample size justification and sample description, the mushroom-picking task, a description of all variables that were used, and a description of the procedure and materials. This research complied with the tenets of the Declaration of Helsinki and adheres to ethical guidelines of the APA Code of Conduct. Informed consent was obtained from each participant. As no profound prior information for the domain of mushroom picking existed, this study was exploratory and thus no preregistration before data collection was possible. Because values of transparency are important to us in the wake of the Open Science movement, we share additional study details, key replication materials, and data and R codes for analysis in the Appendix (Appendix A, Appendix B, Appendix C) and as supplementary material on OSF (newly developed test items, the R-code used for data analysis, raw data with variable descriptions, as well as code used to train the AI model and to generate the explanations can be found under <https://osf.io/xesf7/files/>; note: Data will be uploaded only after acceptance of the manuscript).

4.1. Independent variables

We manipulated the study in a 2×2 between-subjects design by (i) an educational intervention about the AI functionality, and (ii) visual explanations of the AI classification, as indicated in Figure 1. To test the effect of an educational intervention on two levels, one group of participants was given a short introduction to machine learning (see Appendix A.2). This description included high-level concepts about typical image classification networks such as the model used for the study. Explainability was also manipulated on two levels, one group receiving the classification result only, and one group additionally receiving two visual explanations (see Figure 2 and Figure 3). For the explanation methods we chose (i) Grad-CAM as representative for attribution-based techniques, and (ii) an adapted version of ExMatchina as representative for example-based techniques where we show the most similar training image of each predicted class.

4.2. Dependent and control variables

For extensive exploration, we chose a large number of dependent and control variables listed below.

4.2.1. Human classification performance and picking behavior

In our study, we break down the mushroom-picking behavior into two questions: an assessment of a mushroom's edibility, and the final decision of whether to pick it up or not (i.e., take-home decision). Both questions can be answered correctly or incorrectly, and independently from each other. A correct edibility assessment is the identification of an edible mushroom as edible, or of a poisonous mushroom as poisonous; an incorrect assessment is the identification of an

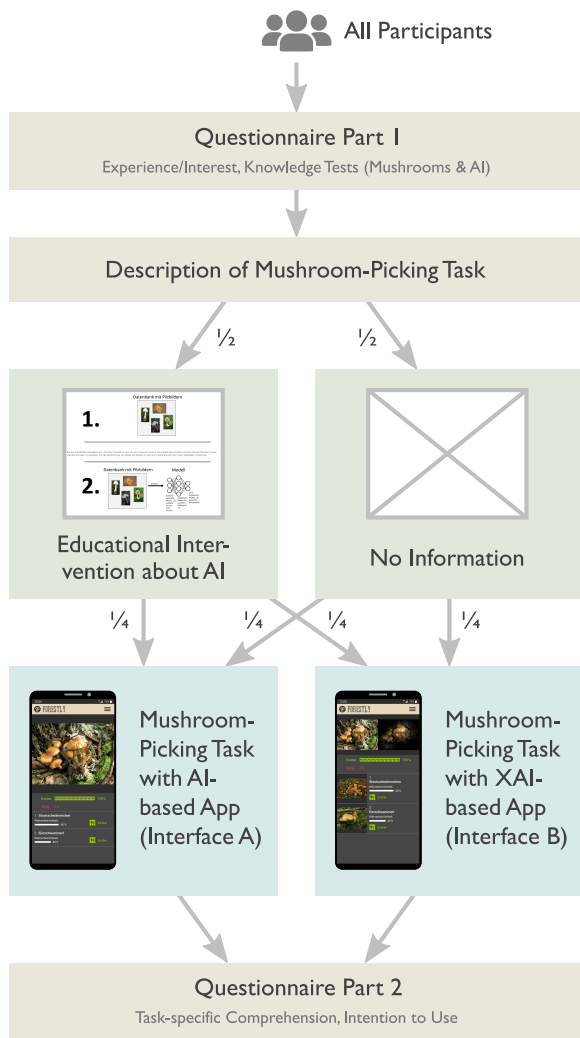


Figure 1: Participant flow in the 2 × 2 study design, including questionnaires, the optional educational intervention, and the mushroom-picking tasks with or without visual explanations. Mushroom images (top to bottom) by Tom Smidth, Eva Lau Olsen, and Frede Scheye.

edible mushroom as poisonous, or of a poisonous mushroom as edible. A correct take-home decision is to pick up an edible mushroom or to leave a poisonous one; and an incorrect take-home decision is to pick up a poisonous mushroom or to leave an edible one. Both of these decisions are made 15 times (as there are 15 pictures presented).

To reflect these different outcomes, we recoded each of the variables—edibility assessment and take-home decision—into performance measures. Obviously, certain incorrect decisions pose a greater risk than others. For example, not picking up an edible mushroom merely leads to missing out on the enjoyment, while incorrectly picking up a poisonous mushroom might have severe health consequences. Thus, it might be interesting to not just understand correct decisions in general but also subsets of decisions with certain characteristics. This is even more important in light of the potentially wrong or misleading predictions made by the AI system, which participants can trust or not.

4.2.2. Trust in AI and self-reported comprehension of AI classifications

In addition to the edibility assessment and the take-home decision, participants were asked to rate how much they trust (“I TRUST this mushroom identification of the AI”) and how much they comprehend (“I UNDERSTAND how the AI arrives at this mushroom classification”) the mushroom identification/classification of the AI on a 5-point Likert scale ranging from “do not agree at all” to “fully agree”.

4.2.3. AI knowledge

Participants’ knowledge on AI technology was measured with 8 items. Each item consisted of a question and a multiple choice answer format with four response alternatives. For each item, only one response was correct (the other three were distractors). All items were evaluated by at least 3 computer scientists in order to ensure content validity of the test. The test can be seen in the supplementary material in German original and English translation (see <https://osf.io/xesf7/files/>).

For example, one question was “A newly developed AI is supposed to detect foxes in images. What is usually the prerequisite for such an AI?” with response alternatives “A) A large collection of fox images, from which similarities and dissimilarities are inferred.”, “B) Few high-quality images of foxes, on which characteristic traits are marked.”, “C) Fixed code written by programmers, that tells the AI exactly how to distinguish foxes from other animals.”, and “D) Knowledge bases on the internet, which the AI accesses actively through search engines”. (Here, option A is correct)

4.2.4. Domain-specific knowledge on mushrooms

The domain-specific knowledge on Austrian mushrooms was assessed with 11 declarative knowledge items. The test was developed in close cooperation with mycologists of the Mycological Working Group Linz at the Biology Centre Linz (Linz, Austria) to ensure content validity. Again, each item was constructed as a multiple choice item with four response options of which only one option was a correct answer.

For example, one question was “The caps of cultivated button mushrooms are...” with response options “A) yellowish”, “B) fuzzy along the edges”, “C) tapered and pointy”, and “D) roundish and hemispherical”. (Here, option D is correct)

4.2.5. Task-specific AI comprehension

In addition to the more general items used to measure the participants’ AI domain knowledge, we created a test to measure how well participants comprehended the mushroom identification AI presented in this study. The test consisted of 5 multiple choice items with 4 response options each. Again, only one option was correct in each case. For example, one question was “What does a percentage of 77.5 % shown in the *Forestry* app for a mushroom species mean?”. The response options for this example were “A) That the AI’s prediction is correct in 77.5 % of the cases”, “B) That the AI is 77.5 % sure that the mushroom belongs to this species”, “C) That 77.5 % of the users trust the decision of the AI”, and “D) That the

AI was able to recognize and categorize 77.5 % of the image area". (Here, option B is correct).

4.2.6. Other measures

For reasons of sample description, we collected data on participants' gender, age, educational level, and their self-reported domain-specific experience and interest (e.g., "Have you ever picked mushrooms?" and "How high would you rate your interest in mushroom picking?"). We also asked participants about their intention to use an AI-based system for mushroom picking before and after the identification tasks ("How high would you rate your willingness to use an app or other software that suggests classifications of a mushroom based on photographs?" and "How high would you rate your willingness to use the *Forestly* app shown in the study in a real mushroom hunt?", rated on a 5-point scale from "very low willingness" to "very high willingness"). Finally, we asked participants to evaluate the presented AI system ("How would you rate the *Forestly* app shown in the study?" rated on two 5-point bipolar scales from "bad" to "good" and from "incomprehensible" to "comprehensible").

As especially online questionnaires are vulnerable to inattentive responding we used attention check items as recommended by Maniaci and Rogge (2014). We used a total of 4 items throughout the questionnaire to screen out inattentive participants to improve data quality (e.g., attention check item placed within the mushroom knowledge test: "Several edible mushrooms can be mistaken for similar-looking poisonous ones. This question checks your attention. Please simply select 'Yellow foot'.", with response options "A) Common ink cap", "B) Pinecone cap", "C) Snowy waxcap", and "D) Yellow foot").

4.3. Mushroom-picking task

In order to study human mushroom picking supported by an AI classifier within an online experiment, we developed a new mushroom-picking task. The task consisted of 15 items in total. Participants were presented with mushroom pictures taken from a database containing mushroom photographs (Danish svampeatlas, 2022). The study participants were asked to imagine going mushroom picking in a forest with the intention to cook a meal from the picked mushrooms (see full description of the imaginary situation in Appendix A.1). In this imaginary scenario, the presented mushroom pictures would be the mushrooms they found. The participants had to decide for each of the mushroom pictures whether they would classify it as edible or poisonous/inedible (see also Appendix A.4), and whether they would ultimately take this mushroom home for consumption or rather leave it in the imaginary forest (see Section 4.2).

For each item, participants were additionally presented with one of two versions of an interface of the fictitious app *Forestly* showing the classification recommendation of a real AI. This classifier takes as input an image of a mushroom and predicts the most likely from a set of 18 species. Details for how we implemented the classifier can be found in Section 4.5.

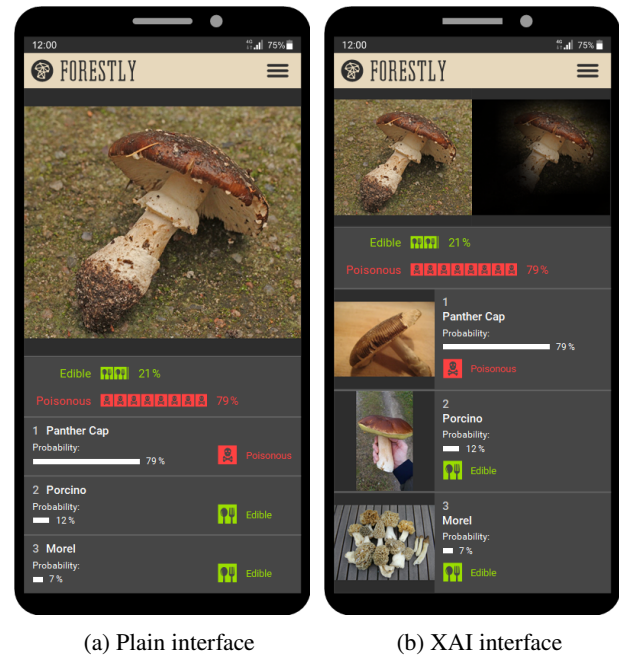


Figure 2: Examples of the two variants of the *Forestly* app interface for a correct prediction, with a "reasonable" explanation in the XAI variant (translated version).

Mushroom images (from top to bottom) by Michael Soniks, Dan Ingemann Jensen, Andreas Roosvall, and Bjørn W. Pedersen.

The classification result was presented to the participants in the form of the top 3 mushroom species as predicted by the AI, along with bars indicating the percentages of certainty for each species. Additionally, the total predicted probability of the mushroom being edible or poisonous/inedible was indicated with a bar chart and numbers. This total probability was determined using the AI's species predictions. Depending on the version of the interface presented to the participants (see manipulations in Section 4.1) the interface additionally included visual explanatory elements: example pictures for each of the top 3 predicted species, and an attribution map indicating important regions in the image. Examples of the *Forestly* interfaces are shown in Figures 2 and 3. For a more detailed description of the explanations, see Section 4.5.

We selected 15 representative images of the training dataset (Danish svampeatlas, 2022) as the final item pool for the mushroom-picking task. This selection of the final set of 15 stimuli was based on the following criteria: (i) Both, edible and poisonous/inedible mushrooms should be included in the set of items (as the value participants assign to an outcome might differ for the various scenarios discussed in Section 4.2.1). (ii) The rating of the AI's certainty of the classification results (ranging from 0 % to 100 % certainty) should vary, as this could be a potential confounding variable, see e.g., Kunze et al. 2019). (iii) The AI classifier should provide false classification recommendations in at least 5 items in order to estimate overtrust (i.e., a person's incorrect assessment of a mushroom's edibility based on an incorrect AI prediction).

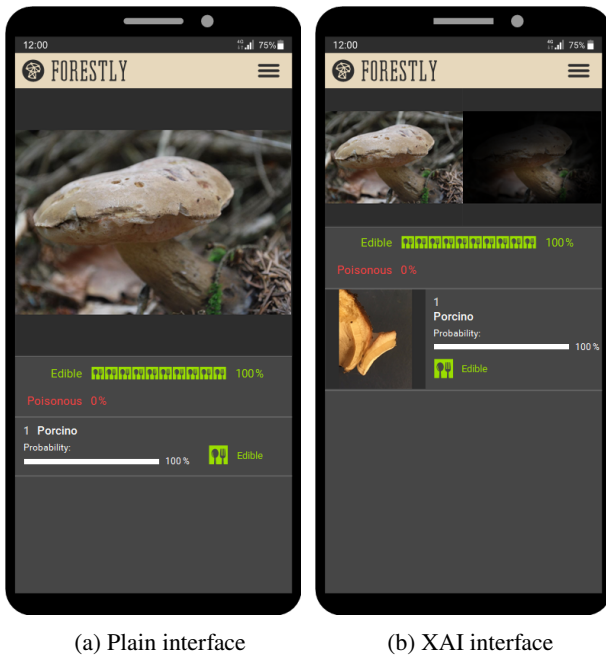


Figure 3: Examples of the two variants of the *Forestly* app interface for a wrong prediction, with a “dubious” example-based explanation in the XAI variant (translated version).

Mushroom images by Poul Olsen (top), and Nanna Heidemann (bottom).

For each task, the mushroom picture (stimulus) and the corresponding image of a fictitious interface, as well as the dichotomous items on edibility classification and picking behavior, the single-item Likert-scale on trust in the system’s recommendation, and a single-item Likert-scale of self-reported recommendation understanding were presented on one page. Each task—and thus each stimulus—was presented on a separate page without the option to return to a previous task. The order of the stimuli was randomized to control for order-effects.

4.4. Procedure

After being screened for eligibility of participation (see Section 4.6), participants were asked to read general information on the purpose, course, and duration of the study, as well as data protection regulations and contact information. We then sought informed consent of every participant. Participation was voluntary and could have been terminated at any point without consequences.

After giving informed consent, participants filled out a questionnaire about their experience and interest in mushroom picking (see Section 4.2.6). Afterwards, participants filled out the knowledge test on Austrian mushrooms and on artificial intelligence (see Sections 4.2.4 and 4.2.3, respectively).

The next section of the study was the AI-assisted mushroom-picking task. Here, participants were first given brief instructions about the imaginary scenario of mushroom picking: “Imagine that you want to cook a tasty mushroom dish for dinner. To this end, you want to collect fresh mushrooms in the forest...” (the full instructions are given in

Appendix A.1). Participants then received brief instructions on the fictional app *Forestly* (which is powered by the AI). In addition, subjects from the experimental group with the educational intervention received further information on how image classification by an AI works (see Section 4.1 on independent variables as well as Appendix A.2 for a screenshot of the educational intervention).

During the mushroom-picking task, participants were presented static screenshots from different versions of the *Forestly* app, depending on whether or not they were in the group that received explanations of the AI’s predictions or not. The instructions on the *Forestly* app mentioned above were tailored to the version shown during the picking task (see Appendix A.3).

After the mushroom-picking task, participants were asked to fill out the task-specific AI comprehension test (see Section 4.2.5). Finally, participants were asked to evaluate the *Forestly* app and answer questions about their intentions to use such an app (see Section 4.2.6).

4.5. (X)AI Implementation

For the purpose of this study, we developed a simple image classification network based on the ResNet50 network (He et al., 2016a) with pre-trained weights from the ImageNet benchmark dataset (Deng et al., 2009). We adapted the model architecture of ResNet50 by removing its top layers (global average pooling and final linear layer) and adding a flatten layer, followed by two dropout and linear layers. We set the dropout rate for both dropout layers to 50 %. The activation function for the first linear layer is a rectified linear unit (ReLU) and the activation function of the final linear layer is softmax. A sketch of the network architecture is given in Figure A4 in the Appendix. We trained the network to classify images of mushrooms into one of 18 mushroom species.

Although we did not conduct an exhaustive hyperparameter search, we tried various settings of (i) different top layers (one, two, and three sets of dropout and linear layers), (ii) the amount of layers (none vs. 15) for which the weights were fixed to the pre-trained ImageNet weights during training, (iii) different input resolutions (224×224 vs. 512×512), and (iv) different pre-trained base models (InceptionV3 by Szegedy et al. (2016); VGG16 by Simonyan and Zisserman (2015); ResNet50V2 by He et al. (2016b); ResNet50 by He et al. (2016a)), to reach satisfying results.

For fine-tuning the model towards our mushroom classification task, we chose the Adam optimization algorithm (Kingma & Ba, 2017) with a categorical cross-entropy loss function, and used a held-out validation dataset for early stopping of the training.

As training data, we acquired mushroom images from various sources. We mostly used images from the FGCvX Fungi Classification Challenge (2018), which is based on the Danish svampeatlas (2022). We randomly split these images into train, test, and validation datasets. Since certain species were underrepresented in our training data, we then added images from Google Images and Wikimedia, so that there

were at least 80 training images per species. This resulted in a total of 3480 training images, 241 test images and 219 validation images ($0.88 : 0.06 : 0.06$). Each mushroom species was represented by between 80 and 450 images. To adjust for this imbalance, each species' contribution to the loss function was weighted accordingly.

Images were preprocessed with the same function that is typically used for the ImageNet dataset (Abadi et al., 2022). During training, data augmentation was performed by randomly varying the rotation, zoom, width and height shift, and horizontal flip of the images.

The accuracy of the chosen classifier was 94 % for the train dataset, 82 % for the validation dataset, and 71 % for the test dataset.³

To obtain estimates of the certainty with which the AI predicts a certain species, we chose Monte Carlo dropout (Gal & Ghahramani, 2016) during inference. We repeat the prediction of a single input 100 times and measure how often each class was predicted. We show the top three predicted classes and their corresponding percentages to users.

As a representative of attribution-based XAI methods, we decided to use Grad-CAM (Selvaraju et al., 2017), since it is a widely used method and passes certain “sanity checks” (Adebayo et al., 2020). Grad-CAM calculates regions in the input image that were particularly important for the AI's prediction. These regions can then be highlighted directly in the image and shown to the user. For our visual explanations, we calculate the Grad-CAM for the top species predicted for an image with respect to the last convolutional layer of the network (with dropout deactivated).

As a representative of example-based XAI methods, we adapted a method proposed by Jeyakumar et al. (2020) that retrieves the training image for a certain species that is closest to the input image in terms of the network's internal representation. We retrieve the closest example image for each of the three most likely species. To determine the similarity between two images, we calculate the Euclidean distance between their representation vectors after the flattening procedure.

The python code for model training and calculations of the explanations can be found in the supplementary material along with the trained model that we used in this study.

4.6. Sample size justification and sample description

Based on the study design, sample size was calculated in an a priori power analysis using G*Power (Faul et al., 2007) for the interaction effect of the two independent variables of educational intervention and explainability. A low to medium effect size was assumed ($f = 0.19$) and the global α -error probability was set to 0.01. To achieve a minimum statistical power of 80 %, a sample size of $N = 327$ was estimated. To

³We would like to note here that it was never our intention to create an actual mushroom-picking assistant to be used in the real world. For such a system, an accuracy of 71 % would be far too low. For this study, however, the low accuracy gives rise to a plenty of incorrect predictions, for which it is interesting from a research perspective how humans use them in their own decision-making process.

be on the safe side, we planned to recruit at least $N = 360$ participants, that is $n = 90$ for each of the 4 groups based on the 2×2 study design. Data collection started on June 14, 2021, and was set to terminate at the end of the day at which we reached the threshold of $N = 360$ participants. All participants were derived from an online panel by a market research company and were paid for participation.

We defined the following inclusion criteria: (i) Since the occurrence of fungal species differs regionally and as there exist cultural differences in the mushroom-picking and consumption habits, only people with main residency in Austria were surveyed. (ii) Only people who eat or would eat mushrooms in general were surveyed. This way, certain error variables can be controlled. (iii) For legal and ethical reasons, only people older than 18 years were allowed to participate.

In total 617 persons started the questionnaire, 33.5 % terminated the questionnaire early or had to be excluded due to low data quality (as detected by certain attention check items; see Section 4.2.6). The total sample size for data analysis was $N = 410$, of which $n = 200$ were in the group with educational intervention compared to $n = 210$ without the intervention, and $n = 202$ participants received visual explanations compared to $n = 208$ without explanations.

From these 410 participants, 213 persons identified as female, 193 as male, 2 persons as diverse (third gender option in German-speaking countries), and 2 persons did not indicate their gender. The mean age of the subjects was $M = 44.58$ years ($SD = 15.29$ years). Only 15.9 % of participants hold a university degree, while most hold a professional, but non-academic educational diploma (53.8 %). When asked about their mushroom-picking experience, 80 % of the participants said that they had been mushroom picking at least once in their life, and of those, 98 % went on a mushroom hunt to actually eat the mushrooms; about 15 % had used an app for the identification of mushrooms in the context of mushroom picking before. The general interest in mushroom picking, however, was rather moderate ($M = 3.04$, $SD = 1.10$, indicated on a 5-point Likert scale ranging from 1 = “very low interest” to 5 = “very high interest”), and the willingness to use a corresponding identification app before study participation was medium to high ($M = 3.69$, $SD = 1.14$, indicated on a 5-point Likert scale ranging from 1 = “very low willingness” to 5 = “very high willingness”).

5. Results

For data analysis, we used the open source statistic software RStudio (R version 4.1.1) (RStudio Team, 2020). The R-Code used for this analysis can be found in the supplementary material.

5.1. Measurement models and item selection

We developed new tests to measure levels of expertise in fields that are likely to influence the results of the study. As mentioned in Section 4.2, we measure participants' prior general knowledge on mushrooms and AI, and we measure the task-specific AI comprehension after using the *Forestly* app. For all new measurement models it is necessary to

Table 1

Values of model fit for the mushroom and AI knowledge tests, and the task-specific AI comprehension test.

	k	df	χ^2	p	CFI	TLI	RMSEA	SRMR	ω
Mushroom knowledge	6	9	17.81	.04	.90	.83	.05	.04	.44
General AI knowledge	5	5	5.42	.37	~ 1.00	.99	.01	.03	.50
Situational AI knowledge	4	2	2.67	.26	.99	.96	.03	.02	.42

Note: k is the number of items in a test, and df is the degrees of freedom.

test the goodness of fit and analyze psychometric properties. To ensure a good model fit and good properties for further testing, we excluded items if they did not fit the model well or if they were too difficult (i.e., items with $M \leq .25$). We chose the weighted least squares mean and variance adjusted (WLSMV) estimator due to the dichotomous nature of the coded items and its superiority to the maximum likelihood (ML) estimator (Beauducel & Herzberg, 2006). Furthermore, we chose cut-off values of $\geq .96$ for the comparative fit index (CFI), and $\leq .05$ for the root mean squared error of approximation (RMSEA). (Yu, 2002).

Table 1 summarizes the results for the test analysis after item exclusion. All tests reached a moderate to good model fit, but low internal consistency. Tables A1, A2, and A3 (all in Appendix B) give an overview of item characteristics.

In contrast to the single-factor measurement models we created for the knowledge tests from Section 5.1, for the mushroom identification tasks we only calculated the sums of correct items per participant. To enable an efficient measurement in future studies, we reduced the number of items from 15 to 10 by excluding items with extreme difficulties (i.e., items that only a few persons answered correctly or items that most persons answered correctly), since these items do not entail much information. For example, one picture of a Chanterelle was correctly identified by 99 % of the participants and 96 % of participants would correctly pick that mushroom. For this item we can assume that there will be not much difference between groups, which makes it not particularly informative for this study. We ensured that at least 4 items were kept that (i) depicted poisonous mushrooms, and (ii) depicted mushrooms with an incorrect AI prediction, since these subsets of items are of special interest in mushroom-picking scenarios. A list of all 15 items and their difficulties is given in Table 2 (note that item selection was performed before conducting further analysis to avoid confirmation bias).

5.2. Effects of educational intervention and explanations

For the following statistical tests, we used a global alpha-level of $\alpha < .05$. Multiple hypotheses were tested and, as this study is exploratory in nature, a large number of tests were performed. Usually, when conducting multiple tests, the family-wise error rate (FWER) is controlled, which means that the probability of at least one Type-I error is adjusted. However, with a large amount of tests, the FWER control might be too restrictive. Benjamini and Hochberg

thus suggest that “a desirable error rate to control may be the expected proportion of errors among the rejected hypotheses” (Benjamini & Hochberg, 1995, p.290), which is termed false discovery rate (FDR). For this statistical analysis we therefore control the FDR instead of the FWER according to the Benjamini–Hochberg procedure (Benjamini & Hochberg, 1995).

To test the main effects of educational intervention and visual explanation on identification, behavioral intentions and task-specific AI comprehension, we used non-parametric Brunner–Munzel tests (Brunner & Munzel, 2000). The effects were tested (i) with all items in the mushroom-picking task, (ii) with a subset of items for which a poisonous mushroom was depicted, and (iii) with a subset of items for which the classification of the AI was wrong (e.g., classifying a mushroom picture as edible while it is in fact poisonous or inedible). Thus, we performed a total of 14 Brunner–Munzel tests.

For the edibility assessment only the visual explanations had a significant effect when considering either all 10 test items ($BM(398.68) = 4.65, p < .001, d = .44, CI_{95} = [.25, .64]$), or the subset of items for which the AI recommended a false classification ($BM(401.73) = 3.48, p = .004, d = .35, CI_{95} = [.15, .54]$). Participants who received the additional explanations by the AI (all 10 items: $M = 5.29, SD = 1.36$; 4 items with incorrect AI: $M = 1.00, SD = .97$) performed better than participants without visual explanations (all 10 items: $M = 4.72, SD = 1.22$; 4 items with incorrect AI: $M = .69, SD = .84$). Tests were not significant for the educational intervention—neither for all test items, nor for items depicting a poisonous mushroom or mushrooms for which the AI classification was incorrect (e.g., for all 10 items: $BM(403.89) = .59, p = .56, d = -.08, CI_{95} = [-.28, .11]$).

When testing the effects of the educational intervention and explainability on participants’ intention to pick the mushroom in this imaginary situation, again only the visual explanation had a statistically significant effect, but not the educational intervention (e.g., for all 10 items: $BM(407.75) = -0.53, p = .77, d = .01, CI_{95} = [-.18, .20]$). Visual explanations had a significant effect only when considering all 10 items ($BM(400.71) = -2.69, p = .03, d = .26, CI_{95} = [.07, .46]$), but not when only poisonous mushrooms or cases of wrong classifications by the AI were considered. Again, participants were more likely to correctly pick or leave mushrooms when the recommendation of the AI system had

Table 2

Mushroom classification items used in the study. *Ed.* is the ground truth for whether the mushroom is edible (🍄) or inedible/poisonous (☠️). *AI prediction* lists the top 3 results of the classification model, with certainty values for each prediction. *AI corr.* indicates whether the AI prediction was correct (✅) or incorrect (❌). *Ed. (AI)* is the probability of edibility as predicted by the AI. M_{pick} and M_{ed} are the mean performance scores for the picking and edibility prediction tasks, respectively, where 0 means that no one answered correctly and 1 means that all answers were correct. For each item, *Incl.* indicates whether the item was used for analysis (✅) or not (❌).

ID	Species	Ed.	AI prediction	AI corr.	Ed. (AI)	M_{pick}	M_{ed}	Incl.
1	Chanterelle	🍄	Chanterelle 100 %	✅	100 %	.99	.96	❌
2	Bitter Bolete	☠️	Porcino 100 %	❌	100 %	.19	.27	❌
3	Bitter Bolete	☠️	Porcino 92 % Bitter Bolete 6 % Sheathed Woodtuft 1 %	❌	94 %	.19	.27	✅
4	Porcino	🍄	Bitter Bolete 46 % Porcino 25 % Autumn Skullcap 17 %	❌	31 %	.16	.13	✅
5	Blusher	🍄	Panther Cap 85 % Blusher 10 % Death Cap 2 %	❌	12 %	.20	.12	✅
6	Autumn Skullcap	☠️	Autumn Skullcap 68 % Chanterelle 26 % Death Cap 5 %	✅	26 %	.98	.99	❌
7	Panther Cap	☠️	Panther Cap 79 % Porcino 12 % Morel 7 %	✅	21 %	.91	.94	✅
8	Porcino	🍄	Porcino 85 % Panther Cap 15 %	✅	85 %	.89	.80	✅
9	Sheathed Woodtuft	🍄	Sheathed Woodtuft 60 % Chanterelle 40 %	✅	100 %	.76	.60	✅
10	Funnel Chanterelle	🍄	Funnel Chanterelle 62 % Jelly Baby 38 %	✅	62 %	.21	.09	❌
11	Field Mushroom	🍄	Field Mushroom 94 % Destroying Angel 4 % Death Cap 2 %	✅	94 %	.84	.71	❌
12	Bitter Bolete	☠️	Bitter Bolete 40 % Death Cap 31 % Porcino 29 %	✅	29 %	.76	.83	✅
13	Amethyst Deceiver	🍄	Amethyst Deceiver 66 % Sheathed Woodtuft 27 % Lilac Bonnet 3 %	✅	94 %	.40	.23	✅
14	Blusher	🍄	Blusher 70 % Death Cap 27 % Porcino 3 %	✅	73 %	.43	.25	✅
15	False Chanterelle	☠️	Chanterelle 100 %	❌	100 %	.30	.38	✅

visual explanations ($M = 4.71$, $SD = 1.23$) compared to not presenting visual explanations ($M = 4.40$, $SD = 1.09$).

None of the manipulations had a significant effect on the task-specific AI comprehension test after the main task. This means that neither participants with visual explanations ($M = 2.34$, $SD = 1.14$) performed differently in the comprehension test compared to participants without visual explanations ($M = 2.35$, $SD = 1.19$), nor did participants

from the group with the educational intervention ($M = 2.34$, $SD = 1.19$) perform differently compared to participants from the group without the intervention ($M = 2.34$, $SD = 1.14$).

5.3. Does domain-specific knowledge or AI knowledge predict task success?

To test the influence of domain-specific knowledge (i.e., mushroom knowledge) and AI knowledge on task success—that is, correctly identifying the mushroom as edible or poisonous and correctly picking an edible mushroom or leaving a poisonous one—we performed two multiple linear regression analyses. The two independent variables educational intervention and explainability (operationalized via visual explanations), and mushroom knowledge and AI knowledge were included as predictors.

However, both the model for edibility classification performance ($F(4, 405) = 5.73, p < .001, R^2 = .05$) and the model for mushroom-picking behavior ($F(4, 405) = 2.43, p = .05, R^2 = .02$) described the dependent variables rather poorly. Neither mushroom knowledge ($\beta = .06, SE = .05, p = .25$) nor AI knowledge ($\beta = .05, SE = .05, p = .30$) seem to predict participants' mushroom identification performance with the AI recommendations. Like the results reported above, this model also reflects the significant influence of explainability ($\beta = .22, SE = .13, p < .001$), but not of the educational intervention ($\beta = -.10, SE = -.04, p = .44$).

Similar results are obtained when using the intention to pick a mushroom as a dependent variable. Neither mushroom knowledge ($\beta = .02, SE = .04, p = .75$) nor AI knowledge ($\beta = .08, SE = .04, p = .11$) seem to have significant influence. Again, explainability ($\beta = .13, SE = .11, p = .007$), but not educational intervention ($\beta = .01, SE = .11, p = .82$) are significant predictors in this model.

5.4. Self-reported trust in the AI system

As reported in Section 4.2.2, participants' trust in, and self-reported comprehension of the AI classification were measured for each item of the mushroom-picking task. Based on the final 10 selected items from the mushroom-picking task, self-reported trust reached a good internal consistency of $\omega = .88$, and an overall acceptable model fit when trust is considered a single-factor model ($\chi^2(35) = 117.15, p < .001, CFI = .95, TLI = .93, RMSEA = .08, SRMR = .04$). Self-reported comprehension reached also good internal consistency of $\omega = .91$ and acceptable model fit ($\chi^2(35) = 181.52, p < .001, CFI = .93, TLI = .91, RMSEA = .10, SRMR = .04$). In a model including both factors, the latent constructs correlated with $r = .74$, but the model fitted to the data only poorly ($\chi^2(169) = 2729.11, p < .001, CFI = .59, TLI = .53, RMSEA = .19, SRMR = .09$).

Multi-group comparisons for visual explanations and educational intervention (corrected based on the Benjamini-Hochberg procedure) showed that only explainability had an effect on self-reported trust ($BM(403) = 3.91, p < .001, d = -.36, CI_{95} = [-.56, -.17]$) and on self-reported comprehension ($BM(404) = 2.49, p = .02, d = -.21, CI_{95} = [-.41, -.02]$). Participants without explanation trusted ($M = 3.64, SD = .74$) and reported to comprehend ($M = 3.57, SD = .87$) the system more compared to participants that received further explanation (trust: $M = 3.36, SD = .81$;

self-reported comprehension: $M = 3.39, SD = .76$). The educational intervention had no statistically significant effect.

These exploratory results could indicate that participants without additional visual explanations also lean more toward overtrust, as they might follow the AI recommendations without critical reflection. We therefore tested in a linear model if trust ratings predicted worse user classification performance for items that the AI incorrectly classified as edible. These are the cases in which trusting the AI recommendations too much can lead to potential harm (rather than missed pleasure). Indeed, the model ($F(1, 408) = 26.92, p < .001, R^2 = .06$) showed that participants reporting higher trust in the AI also were significantly more likely to incorrectly classify a poisonous or inedible mushroom as edible ($\beta = -.24, SE = .03, p < .001$). Furthermore, they would also more likely *pick* such an incorrectly classified mushroom ($F(1, 408) = 9.84, p < .005, R^2 = .02; \beta = -.15, SE = .03, p = .002$).

5.5. Effects of educational intervention and explanation on AI system evaluation

Finally, we also explored main effects of explainability and educational intervention on users' overall evaluative ratings of the system (i.e., the *Forestly* app). Participants who did not receive any visual explanations evaluated the app as better ($M = 4.12, SD = .82$) compared to participants that received the interface with visual explanations ($M = 3.77, SD = 1.03; BM(390) = -3.44, p = .003, d = .38, CI_{95} = [.18, .57]$). The group without visual explanations also reported higher intentions to use the app ($M = 3.58, SD = 1.10$) than the group with explanations ($M = 3.24, SD = 1.23; BM(404) = -2.73, p = .01, d = .29, CI_{95} = [.09, .48]$). Instructions did not have significant effects on evaluation and intention to use.

6. Discussion

In this exploratory experiment we manipulated both (i) the educational intervention about how the AI-based classifier works, with the goal to enable users to better comprehend AI-based technology, and (ii) the explainability of the AI-based system with attribution-based and example-based explanations. In the mushroom-picking use case, we tested the effects of our manipulations on task performance (i.e., a user's assessment of the edibility of a depicted mushroom and their behavioral intention to pick or leave it), and on user comprehension of the AI assistant, user trust, user evaluation and their intention to use. Furthermore, the influences of domain-specific knowledge (i.e., mushroom knowledge) and AI knowledge were measured for exploratory modeling.

The results showed that the educational intervention had no statistically significant effect on any of the dependent variables. This does not mean that educational interventions have generally no effect as we used only a certain implementation of many possible other implementations of "educational interventions". This lack of effect may thus be attributed to the relative shortness of the intervention used in our study, which

might not have led participants to properly understand the AI. Maybe other, longer interventions would be more effective. For example, Körber et al. (2018) showed study participants videos about automated driving that included functionality, sensors, and trajectory planning—a more intense and vivid intervention compared to text and images. Furthermore, the literature on AI literacy suggests the use of educational elements in which users could explore the boundaries of an AI system (e.g., Long & Magerko, 2020).

Explanations of the AI's predictions, in contrast, were found to be effective, leading to a statistically significantly better performance in the mushroom-picking task. Better performance here means a higher accuracy both in assessing a mushroom's edibility and in picking an edible mushroom but leaving an inedible or poisonous one. However, results were clearer for edibility assessment than for actual intentions to pick a mushroom. While edibility assessment may represent a measure of maximal effort, behavioral intentions to actually take mushrooms may also reflect personality traits, as it might represent typical behavior (for maximal effort and typical behavior see Cronbach 1970 as cited by Olderbak and Wilhelm 2020), such as a greater propensity for risk aversion, which could lead to smaller overall effect sizes.

Domain-specific knowledge and AI knowledge—while shown to be key variables for decision-making with technological support—did not affect performance in the mushroom-picking task. Several reasons may account for this null result. First, we had to construct new measurement instruments (i.e., knowledge tests) for a domain-specific knowledge on mushrooms and AI knowledge. Although we carefully constructed the items for the tests (e.g., by consulting a group of mycologists), it may be that the measurement instruments are only of limited validity in this use case. For example, most people are probably not mushroom experts and know only a few common species. However, in the mushroom-picking task they were also confronted with less common ones. While the test might be able to detect differences between laypersons and experts, it might not be able to detect differences in expertise at a higher knowledge level, which is often required for mushroom picking. The lack of evidence might therefore be a consequence of too little variance at this higher level of mushroom knowledge. Similarly, the AI knowledge test might also be of limited validity in this study due to the study design. It is unclear how a higher level of AI knowledge would have affected performance in this task, since the AI-powered app did not allow exploration. Only fixed, predefined pictures were presented, and there was no opportunity for users to test the AI or particular limits of the app. For example, a knowledgeable person might have taken several pictures of the same mushroom from various angles to test the model. Therefore people with greater AI knowledge who know how to test the quality of an AI-based system were not able to apply their knowledge the way they usually would. Effects of domain-specific and AI knowledge may thus have been eliminated simply because of the artificiality of experimental designs with their restrictions, which may well play a role in a natural application setting, however.

Finally, the results show that participants with visual explanations trusted the AI classification significantly less and reported to understand it less. This could mean that the visual explanations prevented overtrust and thus led to more adequate trust. Participants better understood the limits of the AI's performance and seem to have a more adequate understanding of the system's fallibility. This can be seen in more adequate trust ratings—especially for items for which the AI's prediction is incorrect. Participants had better performance with visual explanations, and showed lower trust and lower intentions to use the app. These results contradict those of other studies at a first glance, such as that of Buçinca et al. (2020). This indicates that the effects have a wide variance, which suggests that specific variables moderate such effects and that more complex models are necessary. For example, trust may also be influenced by AI classification error rates (Kenny et al., 2021), of which participants in our study were not informed. For some of our items, the visual explanations clearly indicated that the classification must be flawed, but were this not the case, and a visual explanation did not hint clearly at an error, such explanations could reassure the user that the classification was correct when it was not. Our results are therefore only valid for this composition of task characteristics. Experiments with other implementations of the task, such as other error rates, may yield different results. It is therefore important that studies with other designs are conducted and the influences of various moderators on effect sizes and directions are calculated in subsequent meta-analyses.

6.1. Limitations

As in every study, a number of limitations need to be considered and can be addressed in future work. First, our study considers only a small subset of possible XAI techniques for the visual explanations. Although we tried to include two practically relevant techniques, this provides a limited view on the subject. We sought a task that is relevant to people and has serious implications if a wrong decision is made, but participants might not have been properly motivated to optimize their decisions, as this was just an online study. This could be improved either by offering a bonus reward or by adding a gamification aspect to the study, and a study setup closer to the real situation of mushroom picking in a laboratory or in the field.

Furthermore, the experimental task might have been too restrictive, leading to null results for AI knowledge and our educational intervention. Maybe more knowledgeable users would have gathered more meta-information about the AI training or the quality of the underlying data, or tested the AI with a sample of test images before actual usage in a mushroom hunt. However, the study was restricted, usage was only simulated, and users were not able to test the app on their own. Thus, it might not have been possible for users with better AI knowledge to actually use their knowledge in order to evaluate the quality of the AI's decision outcomes. Future studies will have to (i) allow users to use the *Forestly* app more freely and (ii) additionally apply

qualitative methods of user observation or log file analyses in order to understand how users might use the app differently depending on technology knowledge. Such artificiality in experiments has been criticized both in the literature on comprehension (Moehring et al., 2016) and in the XAI literature, for example, by Ehsan et al. (2021). The latter emphasized that the interactions of humans with AI are socially situated and thus XAI research must take the socio-technical perspective into account, which was echoed by other researchers calling for XAI studies that are closer to real-world contexts (Kenny et al., 2021).

Ghassemi et al. (2021) argued that local explanations are unreliable or give only superficial information. In our study, explanations led to better performances in correctly picking edible mushrooms because the explanation communicated that the AI might be incorrect (e.g., giving questionable example pictures or only highlighting areas that do not show the actual mushroom). However, this need not be the case. Explanation is neither necessary nor sufficient for making a correct decision. According to Ghassemi et al. (2021), such explanations are useful only for model development, but not for individual decisions. As previously mentioned, the results of our study also need to be carefully examined by considering the specific study and material design. For example, effects of explainability could vary greatly depending on the characteristics of the items chosen. Therefore, study characteristics should be varied in future studies depending on different potential moderators, such as error rates. Additionally, trust dynamics and behavioral change should be studied for long-time usage in order to test boundaries and robustness of these effects. However, although the results in this study might be specific and tied to the context and concrete implementations, explanations led to more appropriate trust in this situation, and thus the mushroom-picking task presented here could be used to educate people in exploring the boundaries of AI, and might thus improve general AI literacy. As discussed by Ng et al. (2021), such exploration of AI can then be seen as an educational intervention.

As mentioned before, the null results for AI knowledge and domain-specific knowledge contradicting previous research (e.g., Greiff et al., 2014; Moehring et al., 2016) could be due to inadequate measurement instruments (e.g., they might not have had the power to measure knowledge within the spectrum needed to predict study outcomes; i.e., they were not sensitive at this level of knowledge). Although not shown by the results, we would still argue that domain-specific knowledge is an important aspect for explanations: In order to know whether the AI looked at appropriate parts of the picture, users need domain-knowledge about mushrooms. Thus, future studies need to explore various levels of expertise to test under which circumstances domain-specific knowledge is a driver of effects.

Finally, we must consider limitations of the statistical analyses we chose. While the use of FDR control was reasonable in this exploratory study due to the high number of statistical tests (Benjamini & Hochberg, 1995), FDR control is not a strict control of errors. This means, to test

the robustness of these results, this study must be replicated with stricter error control (e.g., with methods to control the FWER) to confirm significant findings.

6.2. Mushroom picking task: a use case for further research

In this article we presented the use case of mushroom picking with an AI-based assistant system as a means of studying XAI and AI literacy research questions. Mushroom picking represents an interesting use case for this purpose because it represents a decision-making situation of relevance to users due to the high stakes of the domain, availability of the target population, closeness to participants' reality, and public interest (e.g., in European countries such as Austria, Finland, and Sweden). It offers great potential for future research to explore other aspects of XAI and AI literacy.

A first important step could be replication of this study with the same manipulations but with stricter error control. Since it is an easy task to be implemented in online studies, results can be tested for their robustness.

Furthermore, of course other XAI techniques and various combinations of techniques could be tested with this use case. The educational intervention had no effect in this specific study, but this could be different for other implementations.

Another important aspect would be to replicate such studies then in a study situation closer to the real-world context of mushroom picking to show generalizability beyond online behaviors. For this purpose, two approaches could be used: one could perform a laboratory experiment in which users are instructed to scan specific mushrooms in a restricted area; alternatively a field study with limited control could be conducted. In such study contexts, other (socio-technical) dynamics may become visible that are not apparent in restricted online studies (e.g., see studies on socio-technical effects, such as work by Ehsan et al. (2021)).

Furthermore, it is important to confirm the results with other sets of items and characteristics (e.g., different error rates in AI decisions) in order to explore boundary conditions and robustness of effects more deeply.

For more holistic theory development, future studies could also include other relevant variables, such as personality traits (e.g., propensity to trust or risk aversion).

We used a specific pool of participants (e.g., mostly laypersons from Austria). Cross-cultural comparisons and testing of effects of other user characteristics, such as domain-expertise would be interesting.

Finally, mushroom picking is of interest in some countries, such as Austria and Sweden, but might not be of interest in others. However, this use case might share features with other use cases, for instance, because it is from a high-risk domain or has similar distributions of domain-specific knowledge in the population, and thus results could—to some extent—be generalized to other, similar use cases. To test the generalizability of effects, it would therefore be interesting to compare the results with those from other domains that are similar in the most central characteristics.

With this article, we provide a first basis for further research. We have shown that mushroom picking is an interesting use case for XAI that offers several advantages. We have provided first estimates, developed first measurement instruments of AI knowledge and domain-specific knowledge (i.e., mushroom knowledge), and a possible implementation of a mushroom-picking task for simple online experiments for other researchers to build upon in future studies.

6.3. Practical implications

In our study, users tended to rate the AI-powered app more negatively when they received additional visual explanations. We believe that this feedback reflects the relatively poor performance of our AI, which became more apparent to the users through the explanations. This might tempt developers of real-world applications to hold back explanations in favor of receiving more positive initial user feedback. However, in the long run any application will be judged based on the actual performance users achieve, especially when consequences of wrong decisions are as severe as in the context of mushroom classification (e.g., food poisoning after relying on an incorrect AI classification). We thus encourage app developers to consider including explanations, or otherwise indicate potential flaws in their system.

Implementing the educational intervention about AI with a brief text and information graphics had no significant impact on performance, comprehension, or trust calibration in our study. Reasons for this may be manifold, as previously discussed, but assuming that educational measures are ineffective would be wrong. Rather, the study shows that short, simple educational measures via classic text-picture combinations may not be as effective as hoped, and more elaborate interactive interventions are needed—for example, to allow exploration by users.

7. Conclusion

In this exploratory experiment we showed that the newly conceptualized mushroom-picking task is a promising use case for XAI research allowing the exploration of different research questions within a high-risk decision context. We explored the effects of visual explanations on typical usage characteristics such as performance, trust, and understanding. Furthermore, we explored the influence of common AI literacy predictors (i.e., AI knowledge and domain-specific knowledge) on models, where new questions about the role of potential moderators and boundary conditions opened up. In this first study, we investigated mushroom picking in a simple online environment. However, the use case of mushroom picking also offers exciting questions about generalizability to actual collecting behavior in more natural, less restrictive environments. The results and materials presented here can be a first starting point for such an endeavor.

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Appendix A. Detailed Study Information

Appendix A.1. Introductory text for the mushroom-picking task

The following text was presented as an introduction to the mushroom-picking task. The description was illustrated with copyrighted pictures.

Imagine that you want to cook yourself a tasty mushroom dish for dinner. For this you want to pick fresh mushrooms in the forest, from which you will prepare this meal with the intention of eating it. Of course, it is important that you use only edible mushrooms and not inedible or even poisonous ones. To support you identifying mushrooms, an app called *Forestly* has been developed that uses artificial intelligence to analyze photos of mushrooms and decide (i) what type of mushroom it might be, and (ii) whether it is edible or inedible/poisonous. In the following, you will be shown a series of photos of mushrooms taken while being on the mushroom hunt. Your task is then to decide for each photo whether you would classify the mushroom shown in the photo as edible or inedible/poisonous, and whether you would take and eat that mushroom. As a possible decision aid, you will see the decision of the *Forestly* app, which analyzes the photo using artificial intelligence. For further development of the app, each decision also asks questions about how much you trust this app.

Appendix A.2. Educational intervention

Before the study starts, here are a few brief facts about the "Forestly" app

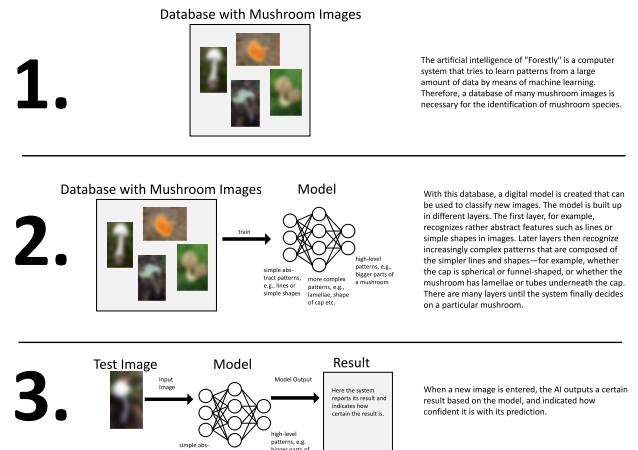


Figure A1: Educational intervention, describing briefly the requirements for, and inner workings of, a typical convolutional neural network (translated version).

Appendix A.3. Forestly App Description

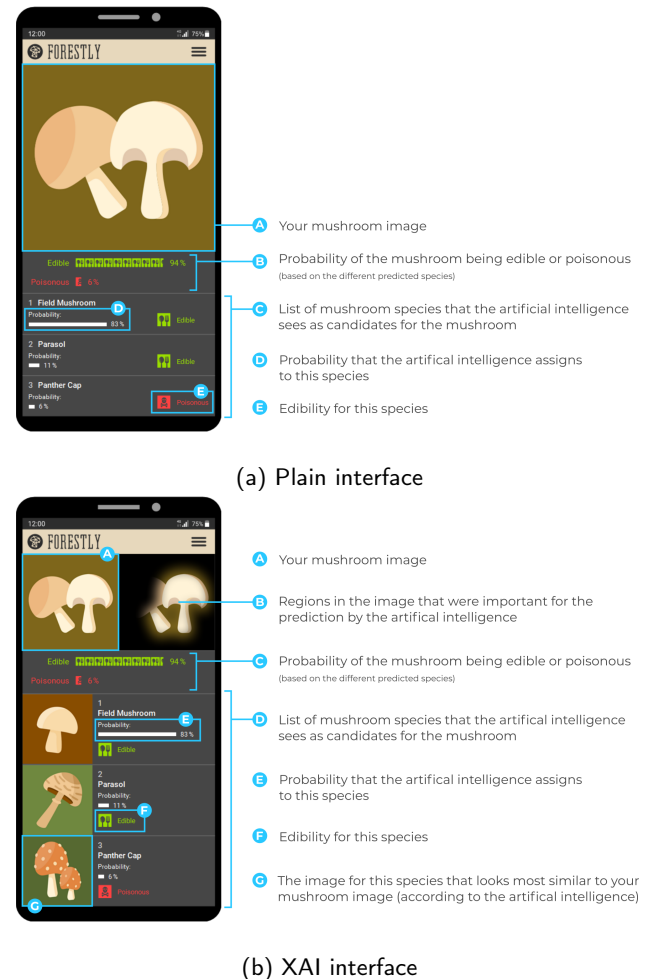


Figure A2: Information about how the *Forestly* app displays the AI's prediction for two interface variants (translated versions).

Appendix A.4. User interface for edibility assessment

How would you classify this mushroom yourself?

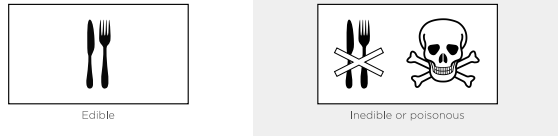


Figure A3: User interface for the edibility assessment task (translated version).

Appendix B. Item characteristics for the mushroom knowledge, AI knowledge, and task-specific AI comprehension test

Table A1

Difficulties, standard deviations and item-total correlations for the mushroom knowledge items.

Item	<i>M</i>	<i>SD</i>	<i>r_{it}</i>
Agaricus	.93	.26	.24
Location	.68	.47	-.03
Imleria badia	.31	.46	.40
Funnel form	.64	.48	.15
Mushroom type	.43	.50	.40
Porcino location	.43	.50	.42
Sweetbread mushroom	.22	.42	-.01
Sheathed woodtuft	.23	.42	.03
Parasol mushroom	.42	.49	.15
Porcino	.27	.44	.37
Poison	.16	.37	.14

Note: **Bold items** were included in the measurement model.

Table A2

Difficulties, standard deviations and item-total correlations for the AI knowledge items.

Item	<i>M</i>	<i>SD</i>	<i>r_{it}</i>
Fox	.53	.50	.44
Basis	.27	.45	.48
Improving	.64	.48	.34
Racism	.67	.47	.23
Statements	.26	.44	.10
Comprehension	.77	.42	.19
Tasks	.58	.49	.18
Logistics	.61	.49	.41

Note: **Bold items** were included in the measurement model.

Table A3

Difficulties, standard deviations and item-total correlations for the task-specific AI comprehension items.

Item	<i>M</i>	<i>SD</i>	<i>r_{it}</i>
Percent	.70	.46	.29
Error	.43	.50	.45
Class	.55	.50	.33
Expert	.66	.47	.37
Single class	.21	.41	-.04

Note: **Bold items** were included in the measurement model.

Appendix C. Network architecture

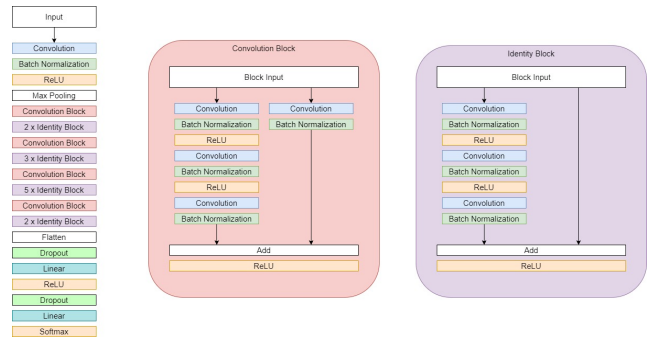


Figure A4: Network Architecture of the classifier used to generate the data for the user study. The Architecture is based on ResNet50 (He et al., 2016a), with the last layers being replaced by custom linear layers. The architecture (left) consists of several **Convolution** (center) and **Identity Blocks** (right).